eflips-x: E-Bus Simulation

Ludger Heide and Shuyao Guo

Solution

Import

GTFS public data and **operator's internal data** formats are supported.

Block Building
Depot-Block Assignment
Terminus Charger Placement
Consumption Simulation
Depot Simulation
Smart Charging

All the parts needed to go from customer-facing schedules to a **simulated electrified operation** are present.

Multiple vehicle **types, depots, days** and charging **strategies** are supported.

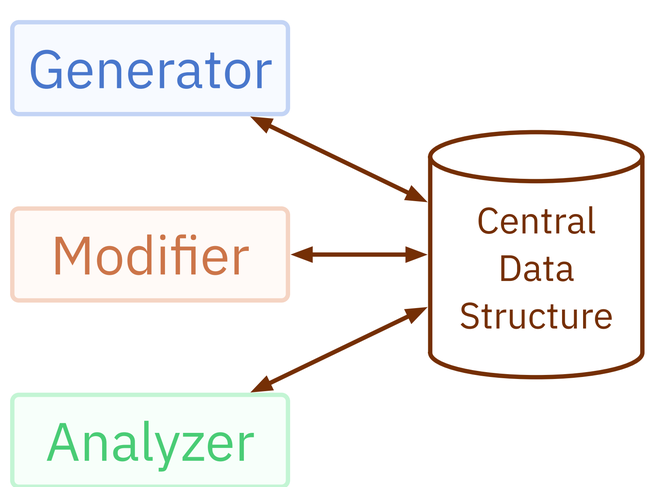
The approach is **scalable** from individual bus lines to the **largest bus network in Germany**

Operational Analysis
Cost Analysis
Environmental Analysis

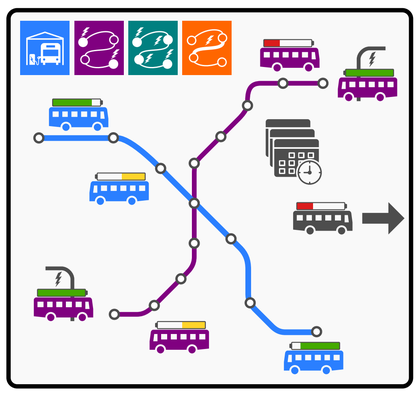
Direct analysis provides vehicle counts, battery exhaustion, charging behavior...

By enriching the direct results with **cost** and **environmental** factors, **total cost of ownership** (TCO) analyses and **life cycle assessments** (LCA) can be conducted.

The modular architecture utilizes **widely-used** programming approaches (Python, SQL) and adheres to **best practices**.



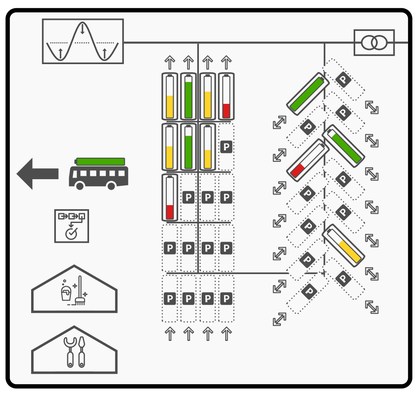
Previous Work



Enroute Simulation

- Energy Consumption
- Terminus Charging
- Network Planning

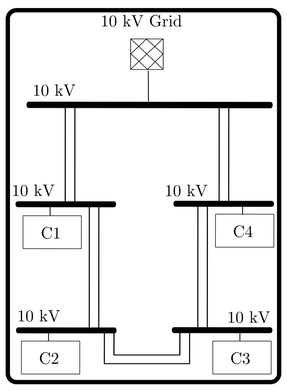
Dominic Jefferies, 2021



Depot Simulation

- Depot Processes
- Parking Layout
- Charging Optimization

Enrico Lauth, 2021



Grid Simulation

- Energy Consumption
- Power Grid Impact
- Fleet Composition

Amra Jahic, 2024

Research Gaps

A **lack of open-source code** and **well-documented** methodologies. Each scientist needs to **reimplement all parts** of the simulation framework, **even if their research focus is only one of them.**

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There is an abundance of open transit data. However, it **lacks blocks**, depot **locations** and **energy consumption information.**

It is almost **impossible** to compare the result quality of different approaches. Reproduction of other author's work and incremental improvement is **severely hampered** by the **unavailability** of **data** and **code.**

Impact

100% electrification case study for the Berlin bus network: ***See companion poster***

The WeBus platform **democratizes** bus electrification by making simulation **accessible to all.**

In the Mobility2Grid **research campus,** eflips-x **simulates** the electric trucks of **BLG Logistics.**

All code is available as **Free Software under the AGPLv3 license.**

Methods for Product Development and Mechatronics

TECHNISCHE UNIVERSITÄT BERLIN